# Diabetes Prediction Using Machine Learning: A Data Science Report

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Project Type: End-to-End Data Science & Machine Learning Project

## 1. Introduction

Diabetes is one of the most common chronic diseases worldwide and can lead to severe health complications if not detected early. Machine learning provides an opportunity to identify individuals who may be at risk by analysing demographic and clinical information. This project applies data science techniques to develop a predictive model capable of classifying whether a patient has diabetes.

The project follows the complete Data Science lifecycle, including data collection, data cleaning, exploratory data analysis, machine learning, model evaluation, explainable AI, and deployment.

## 2. Objective

The objective of this project was to develop and compare several machine learning models to predict whether a patient has diabetes.

The project also aimed to:

- Clean and prepare the dataset
- Explore relationships between variables
- Build multiple classification models
- Compare model performance
- Explain predictions using SHAP
- Save the best-performing model for deployment

## 3. Dataset Description

The dataset contains approximately 100,000 patient records with demographic and medical information.

The variables include:

- Gender
- Age
- Hypertension
- Heart Disease
- Smoking History
- Body Mass Index (BMI)

- HbA1c Level
- Blood Glucose Level
- Diabetes (Target Variable)

After data cleaning, duplicate records were removed, leaving 96,146 records for analysis.

**4. Data Cleaning and Preprocessing**

Several preprocessing techniques were applied before model training.

These included:

- Removing duplicate records
- Checking missing values
- Encoding categorical variables
- Scaling numerical variables using StandardScaler
- Splitting the dataset into training and testing sets using stratified sampling
- Balancing the training data using SMOTE

These steps ensured that the data was suitable for machine learning while reducing bias caused by class imbalance.

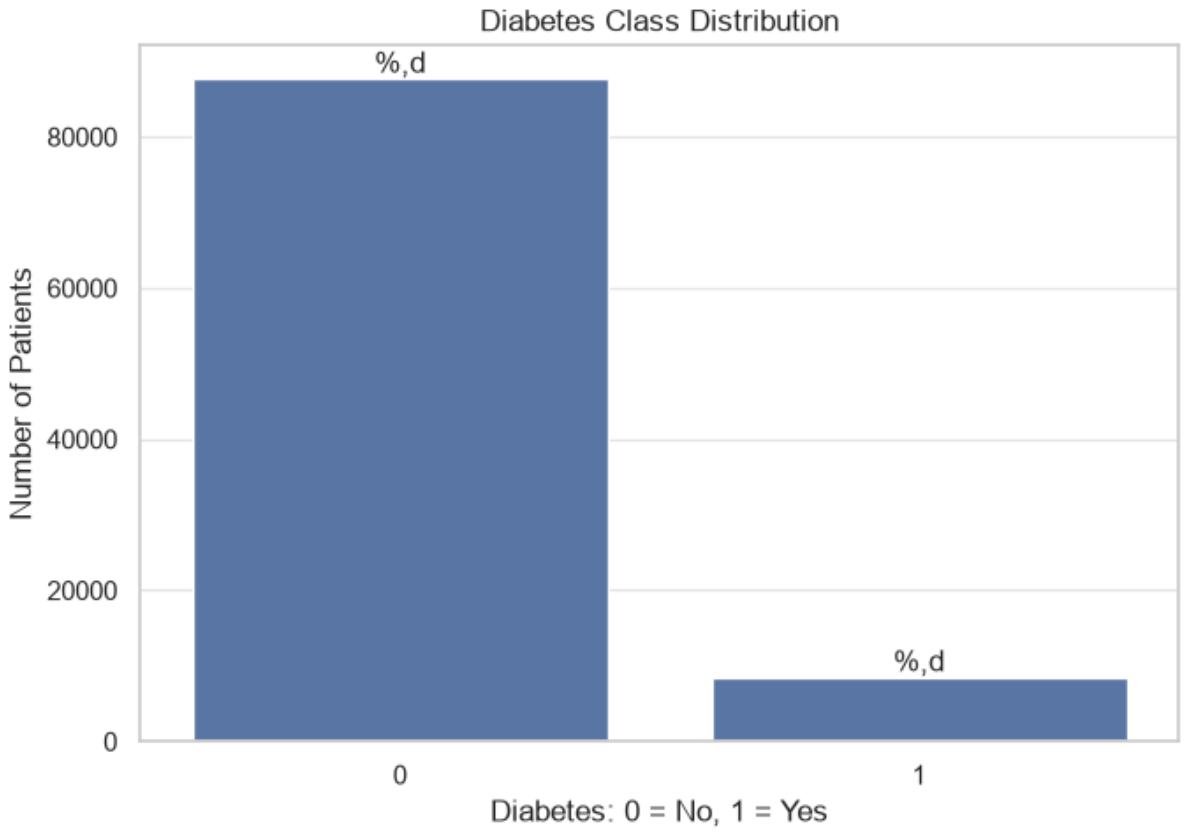
**5. Exploratory Data Analysis (EDA)**

Exploratory Data Analysis was performed to better understand the dataset.

The following analyses were completed:

- Class distribution
- Correlation analysis
- Feature distributions
- Outlier detection
- Numerical feature analysis

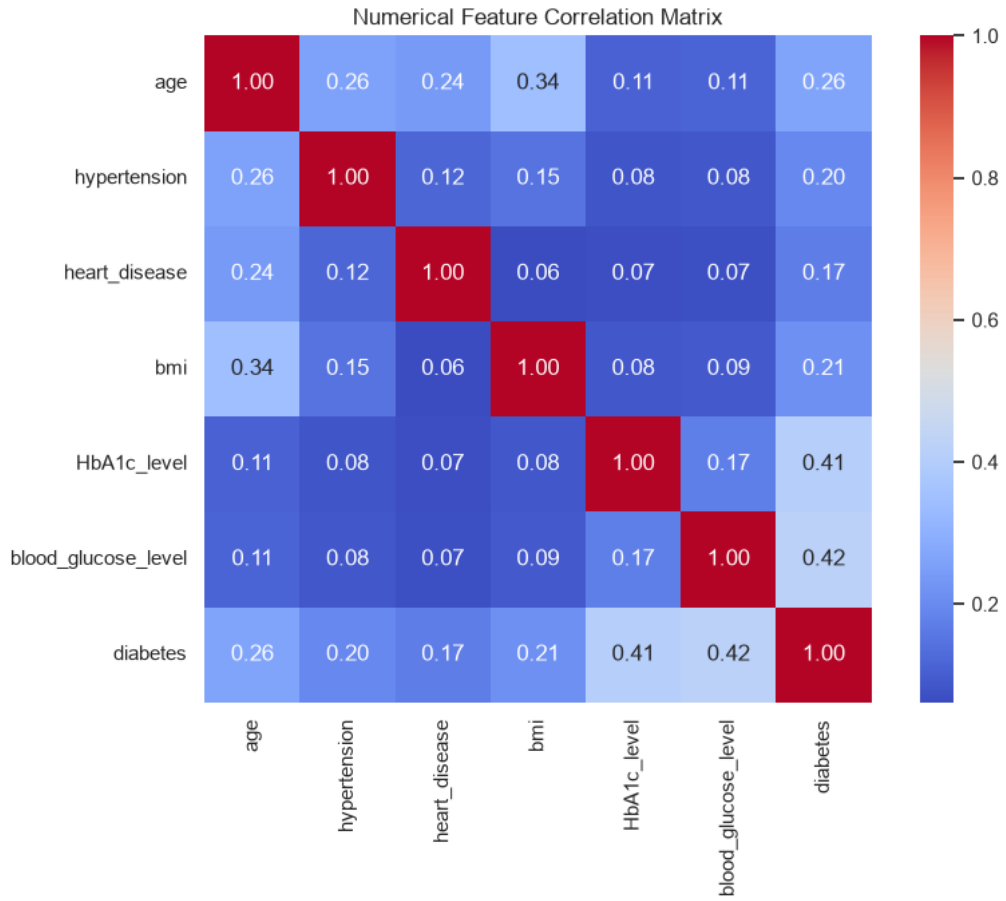
Figure 1: Class Distribution




Interpretation

The dataset is imbalanced, with significantly more non-diabetic patients than diabetic patients. This justified the use of SMOTE during model training.

Figure 2: Correlation Heatmap




Interpretation

The heatmap shows that HbA1c Level and Blood Glucose Level have the strongest relationship with diabetes.

**6. Machine Learning Models**

Four classification algorithms were trained and compared.

- Logistic Regression
- Random Forest
- Support Vector Machine (SVC)
- XGBoost

Each model used the same preprocessing pipeline to ensure a fair comparison.

**7. Model Evaluation**

The models were evaluated using:

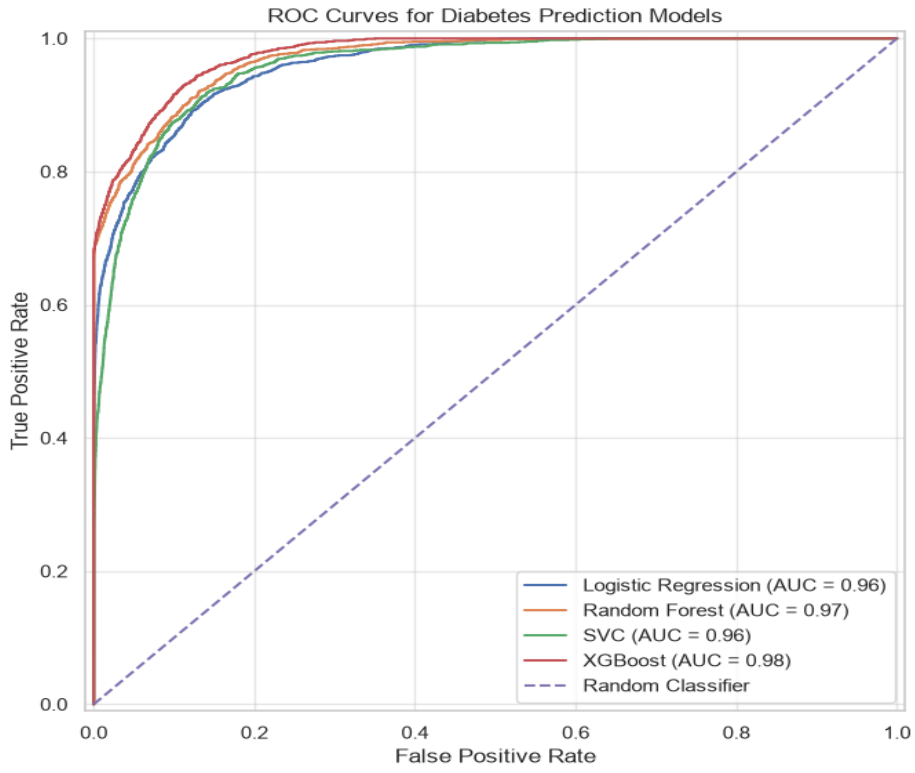
- Accuracy
- Precision

- Recall
- F1 Score
- ROC-AUC
- Confusion Matrix
- Cross Validation

The best-performing model was XGBoost.

| Metric | Score |
|---|---|
| Accuracy | 96.85% |
| Precision | 89.44% |
| Recall | 72.88% |
| F1 Score | 80.31% |
| ROC-AUC | 97.64% |

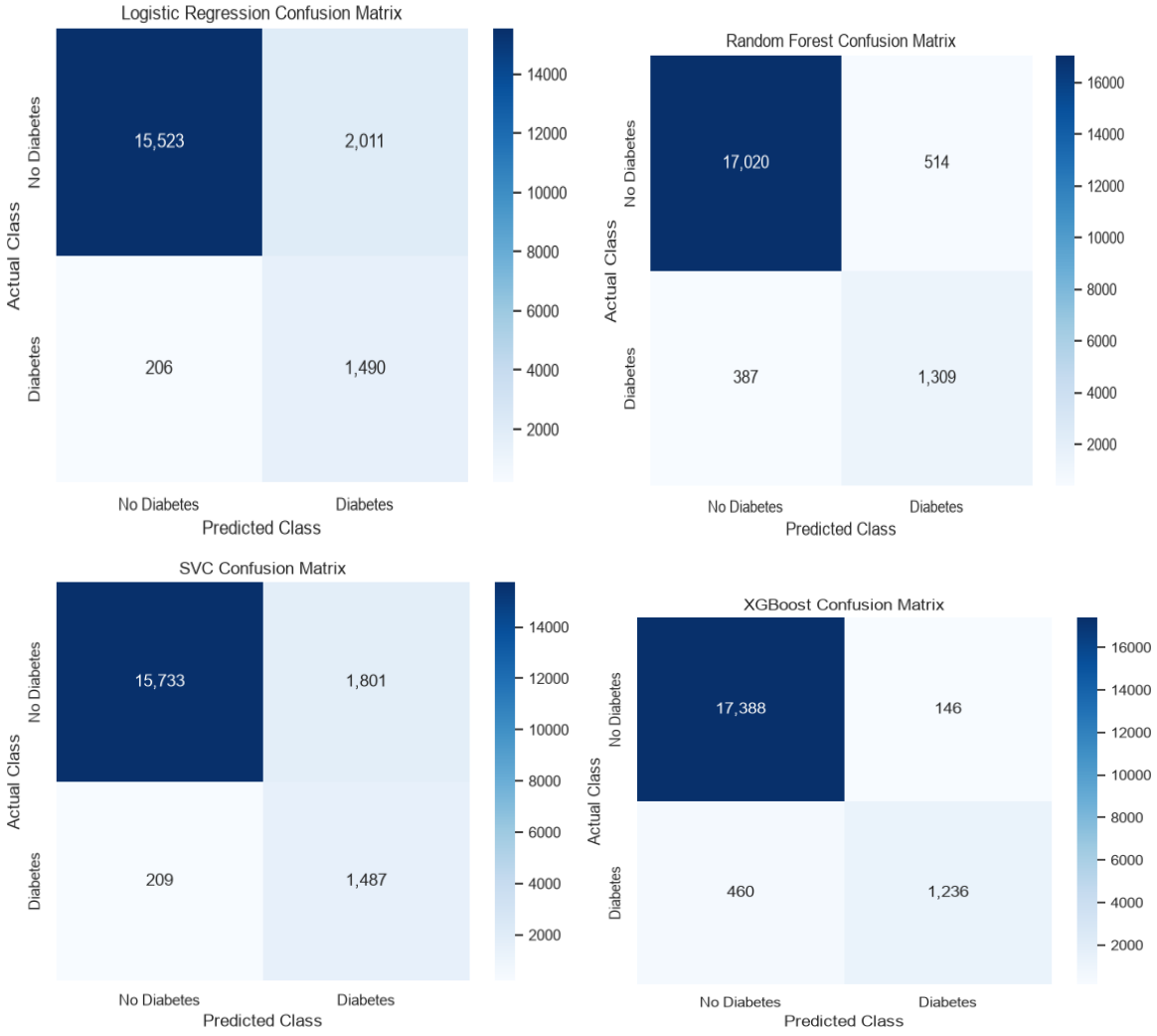
Figure 3: ROC Curve




Interpretation

The ROC curve demonstrates that XGBoost achieved the highest ROC-AUC value, indicating excellent discrimination between diabetic and non-diabetic patients.

Figure 4: Confusion Matrix




Interpretation

The confusion matrix shows that the model correctly classified most patients while maintaining a relatively low number of false positive and false negative predictions.

**8. Explainable Artificial Intelligence (XAI)**

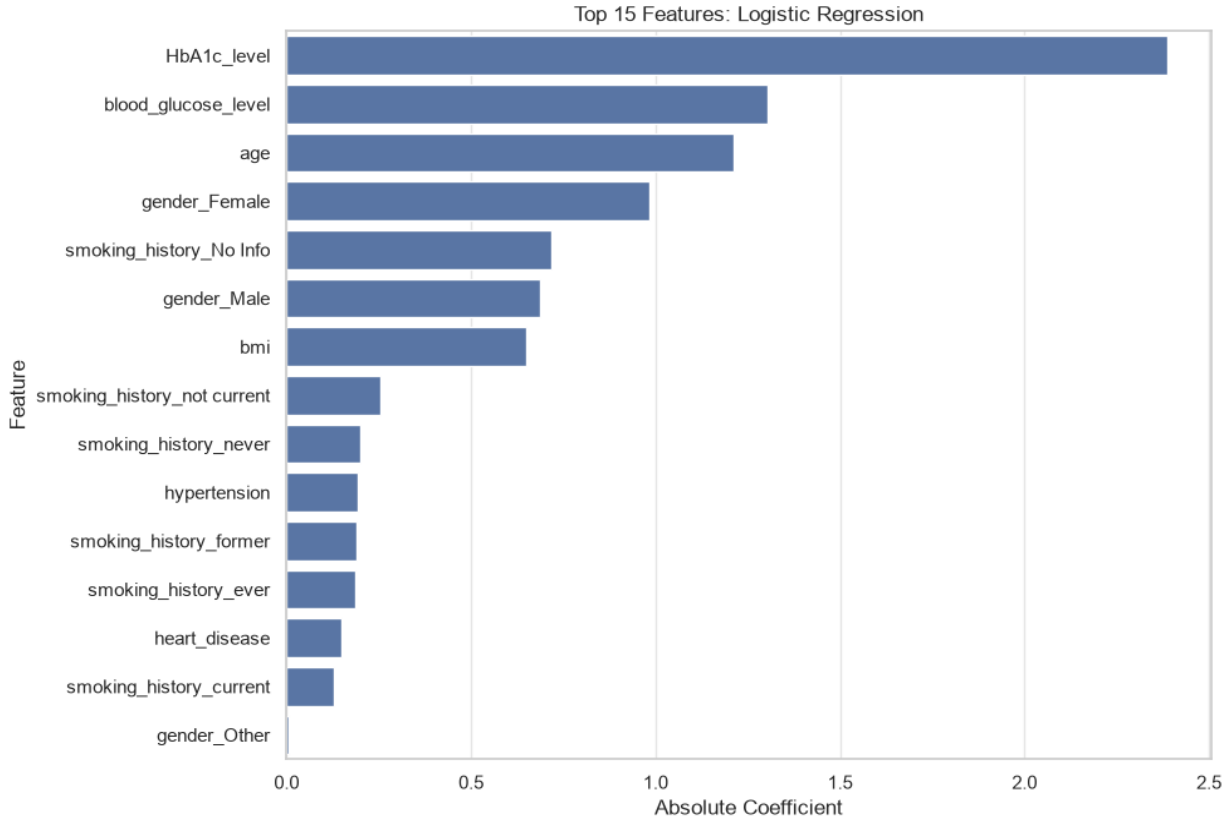
To improve model transparency, SHAP (SHapley Additive Explanations) was used.

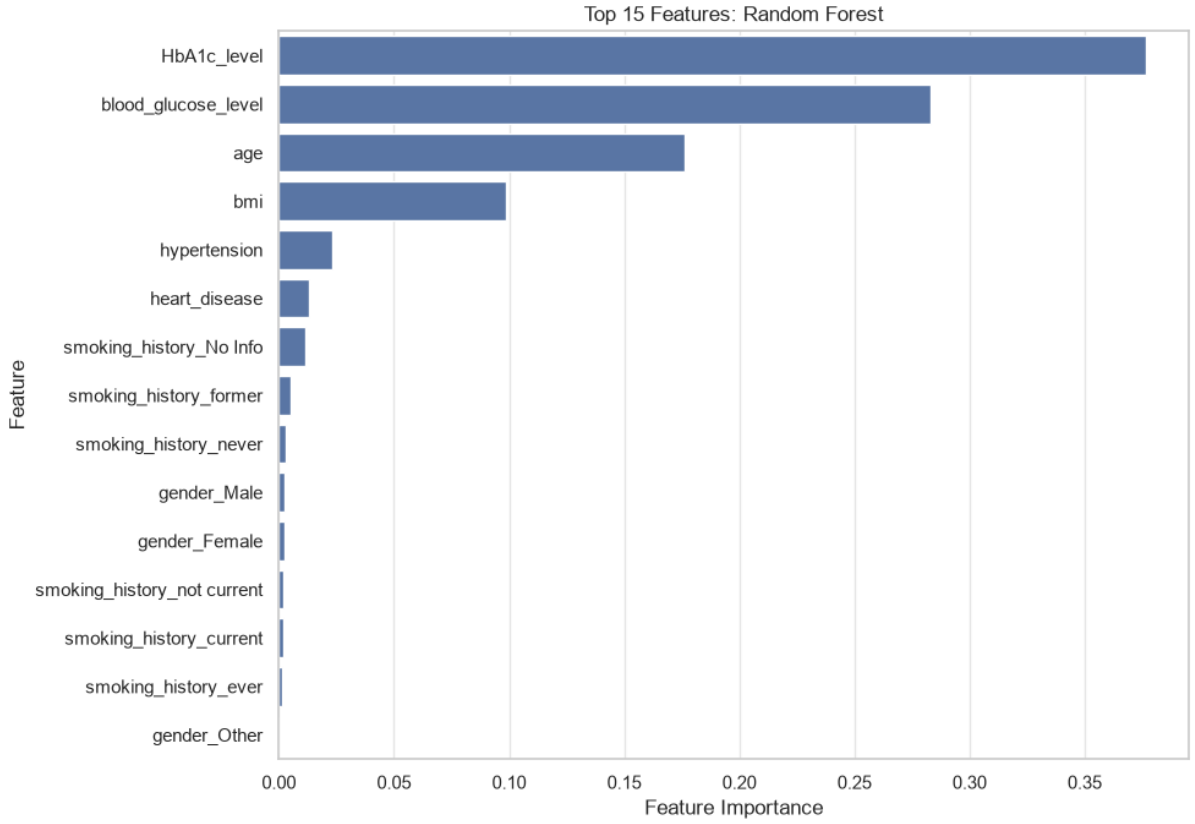
Feature importance analysis identified the most influential predictors as:

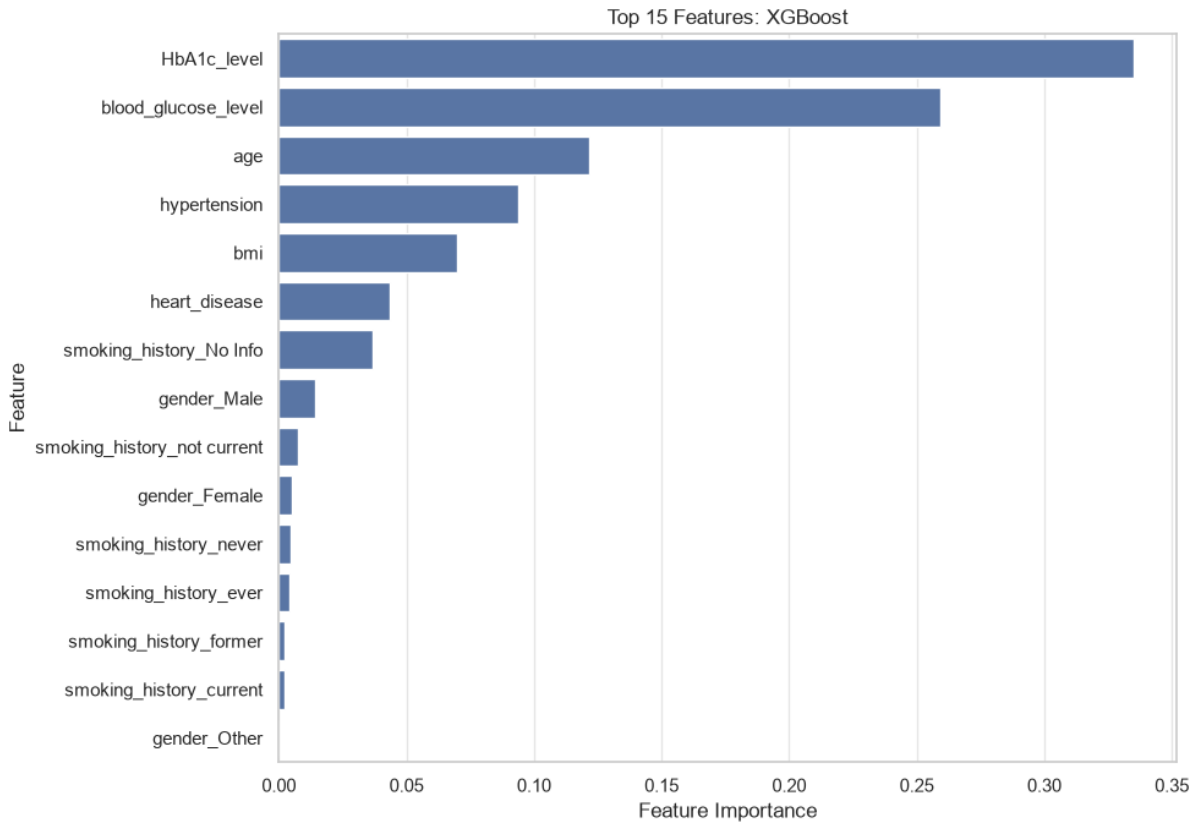
- HbA1c Level
- Blood Glucose Level
- Age
- Hypertension
- BMI

- Heart Disease

Figure 5: Feature Importance



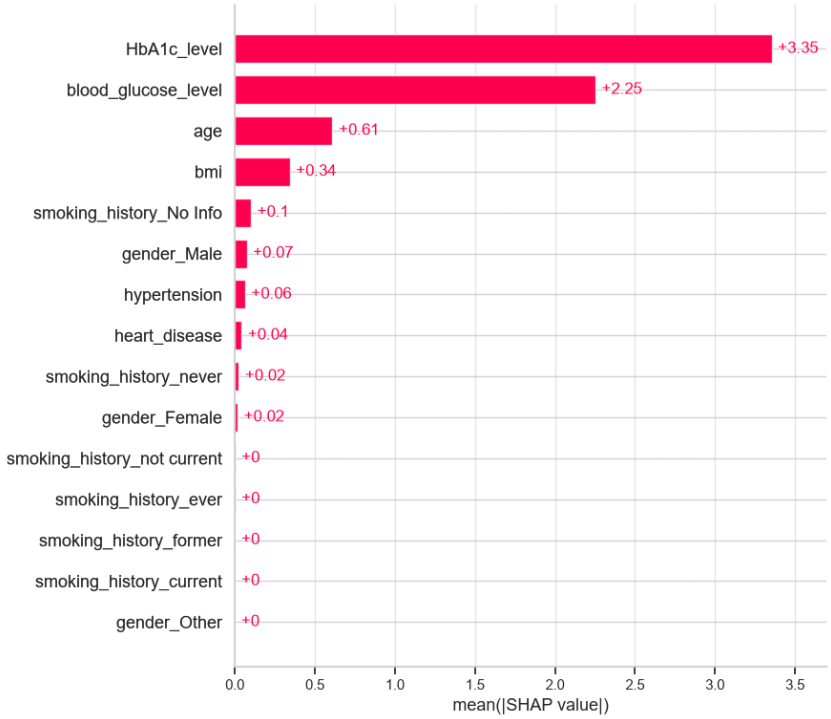



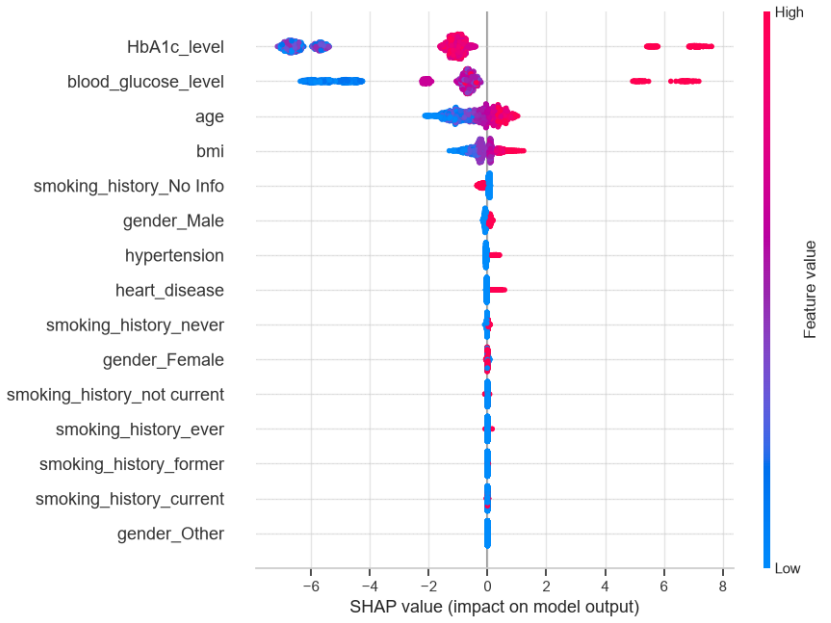


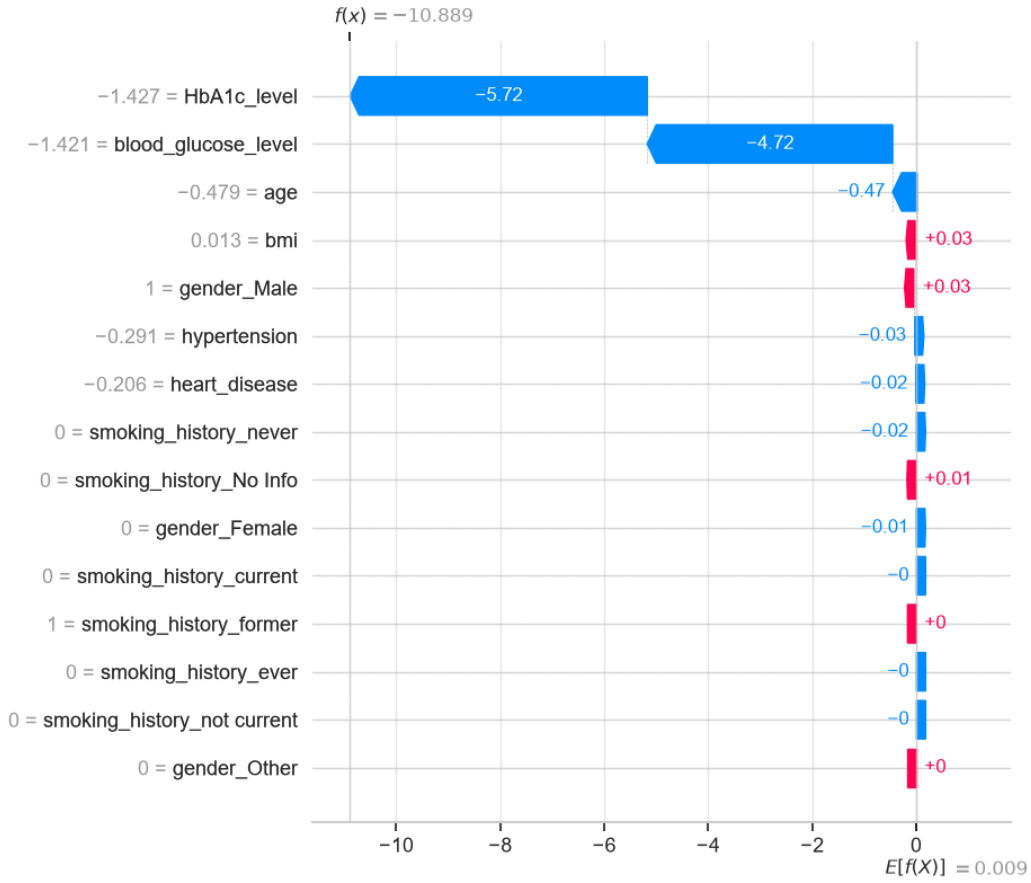

Interpretation

HbA1c Level and Blood Glucose Level contributed the most to the model's predictions, indicating that these variables are the strongest indicators of diabetes within the dataset.

Figure 6: SHAP Summary Plot


HbA1c_level
blood_glucose_level
age
bmi
smoking_history_No Info
gender_Male
hypertension
heart_disease
smoking_history_never
gender_Female
smoking_history_not current
smoking_history_ever
smoking_history_former
smoking_history_current
gender_Other
+3.35
+2.25
+0.61
+0.34
+0.1
+0.07
+0.06
+0.04
+0.02
+0.02
+0
+0
+0
+0
+0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
mean(|SHAP value|)


HbA1c_level
blood_glucose_level
age
bmi
smoking_history_No Info
gender_Male
hypertension
heart_disease
smoking_history_never
gender_Female
smoking_history_not current
smoking_history_ever
smoking_history_former
smoking_history_current
gender_Other
−6
−4
−2
0
2
4
6
8
SHAP value (impact on model output)
High
Feature value
Low




Interpretation

The SHAP summary plot explains how each feature contributes to the prediction, increasing transparency and helping users understand the behaviour of the model.

**9. Deployment**

The trained preprocessing pipeline and XGBoost model were saved using Joblib.

This allows the model to:

- Predict diabetes for new patient records
- Be integrated into future applications
- Produce reproducible results

**10. Ethical Considerations**

Although the model achieved high predictive performance, it should not be used as a replacement for professional medical diagnosis.

Important considerations include:

- Dataset bias

- Patient privacy
- Model fairness
- Clinical validation
- Human oversight

The model is intended for educational and research purposes only.

**11. Conclusion**

This project successfully demonstrates the complete Data Science workflow, from data preparation to model deployment.

Among the four algorithms tested, XGBoost produced the strongest overall performance with an accuracy of 96.85% and a ROC-AUC score of 97.64%.

The project highlights the importance of combining data preprocessing, exploratory data analysis, machine learning, explainable AI, and ethical considerations when developing predictive healthcare models.

Overall, this project showcases practical skills in Data Science, Machine Learning, and Explainable AI, making it a strong portfolio project for healthcare analytics.